

How to Evaluate Semantic Fidelity Across an AI Workflow

A Diagnostic Method for Summaries, Retrieval, and Agents

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Series Context

This paper applies the definitions in *What Is Semantic Fidelity*, *When Accuracy Is Not Enough*, and *The Compression Paradox in AI*. Semantic fidelity concerns the preservation of meaning across transformation. Semantic drift is gradual change in that meaning. Reality Drift is a broader system condition in which representations continue organizing action while losing corrective contact with what they represent.

Shared Illustrative Case

The following report is fictional and isolates the mechanism. A source report says:

“In this small observational sample, Clinic A had fewer repeat visits after introducing remote follow-up. Patients also received additional staffing support. The report cannot determine which change produced the difference, and the result should not be generalized to other clinics without further study.”

A later summary says:

“Remote follow-up reduced repeat visits at Clinic A.”

This paper examines one point at which the later sentence can be produced, accepted, or reused.

Core Claim

An evaluation must inspect the transformations between a referent and an output, because a correct-looking final answer can conceal where meaning changed.

The clinic example can pass several narrow checks. The retrieved report is relevant. The assistant mentions the correct clinic and outcome. The response follows the requested format. An automated judge may see a plausible answer with a citation. None of those checks establishes that the report’s uncertainty about causation survived into the recommendation.

The first step is to name the referent and intended use. Is the system representing a report, a real-world condition, or a user’s decision? The report itself does not prove what will happen at another clinic. The evaluation should therefore state what a faithful answer may conclude and what must

remain open. Without that reference, “*good answer*” easily becomes a judgment of style or task completion.

Next, record the source’s governing relationships before examining the output. In the clinic case these are the small observational sample, the two simultaneous changes, the observed decline, the unresolved cause, and the limit on generalization. Record which of these must survive for the task. This prevents an evaluator from reading a polished output first and unconsciously treating its framing as the source’s framing.

Then trace the handoffs. The original report becomes indexed passages; passages become retrieved context; context becomes an answer; an answer may become a brief, instruction, or input to another agent. At each handoff, ask what was selected, what was omitted, what was inferred, and whether the next step could recover the missing material. A multi-agent workflow can complete every assigned subtask while its final product departs from the initial intent. Local completion is useful evidence, but the whole chain also needs comparison with the original task and source.

Fourth, compare implications rather than matching only words. Does the output preserve who or what the finding applies to, the direction of the evidence, the level of certainty, and the conditions under which a decision would be justified? The sentence “*Remote follow-up reduced repeat visits*” shares many words with the report. It changes the causal implication. A similarity score or correct citation will not by itself reveal that change.

Fifth, test correction. Present the omitted staffing detail and the report’s causal limitation to the system or reviewer. Does the answer change? Can the final reader see the source passage that resists the claim? If an agent already used the wrong answer in a plan, can the plan be revised? Semantic fidelity is partly a property of an output, but in a deployed workflow it also depends on whether contrary source material can interrupt subsequent action.

Finally, observe deployment over time. A benchmark may measure an answer on a fixed question while real use involves ambiguous requests, changed contexts, follow-up questions, memory, and downstream reuse. Evaluation should sample these conditions and retain a path back to the originating evidence. This does not require collapsing all qualities into one score. It calls for separate findings about retrieval, interpretation, attribution, preservation of intent, and correction.

Applied to the clinic case, the diagnosis is specific. Retrieval succeeded. A statement about declining visits survived. The causal limit did not. The answer generalized beyond the report and converted the result into advice. A later evaluation that accepted the citation as proof of grounding missed that transformation. The correction is equally specific: restore the staffing detail and uncertainty, narrow the claim, and reconsider the recommendation for the new clinic.

Diagnostic Question

What changed between source, intermediate representations, and final use, and which step can still be corrected by the original evidence?

Connection to Reality Drift

Evaluation blindness allows a local fidelity failure to remain useful, repeatable, and rewarded. When that pattern persists across a system, operational success can continue while its representations become less answerable to the conditions they claim to describe. The absence of visible failure is not proof of alignment.